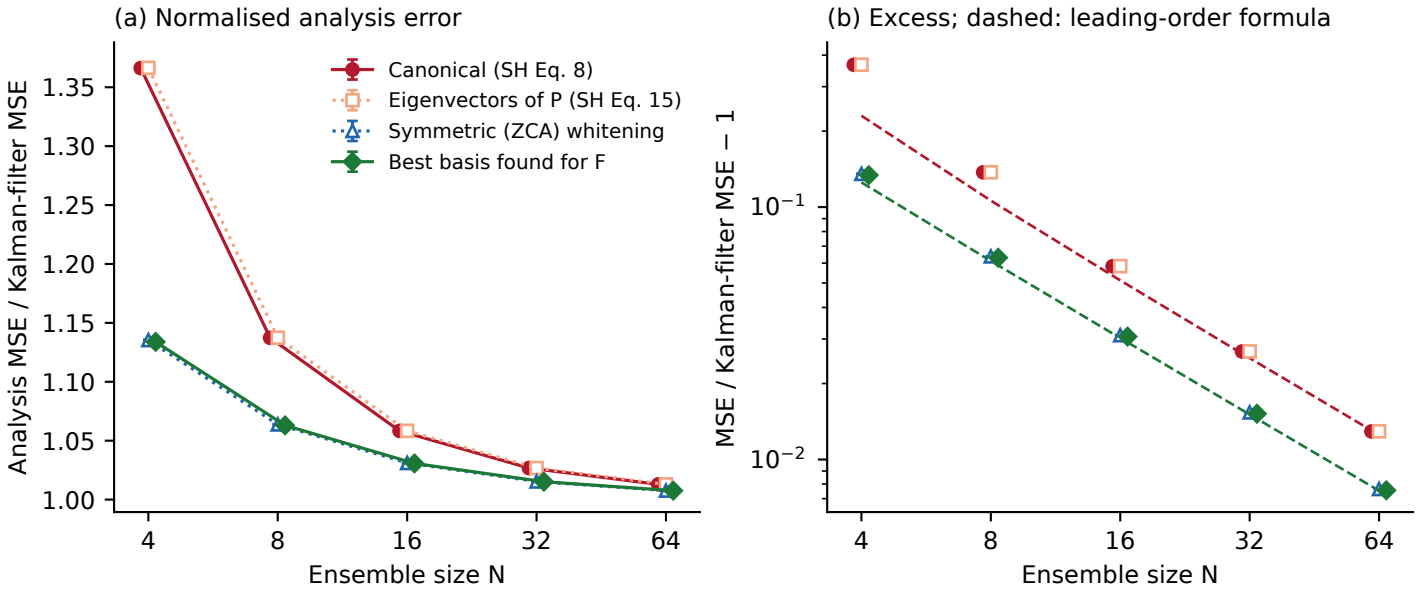
(a) Normalised analysis error
Analysis MSE / Kalman-filter MSE
1.35
1.30
1.25
1.20
1.15
1.10
1.05
1.00
Canonical (SH Eq. 8)
Eigenvectors of P (SH Eq. 15)
Symmetric (ZCA) whitening
Best basis found for F
4
8
16
32
64
Ensemble size N
(b) Excess; dashed: leading-order formula
MSE / Kalman-filter MSE − 1
10^{-1}
10^{-2}
4
8
16
32
64
Ensemble size N